

Atomic MIMO Receivers

Reproduction of Cui et al. "Towards Atomic MIMO Receivers"
Formulae, Methods and Results
(IEEE JSAC 43(3), March 2025)

and an exact-model ULA channel-estimation study

Track A — detection on the Cui 3GPP TR 38.901 channel

Track B — channel estimation on a geometric ULA channel

*Every equation in this report was transcribed from the
implementation and verified numerically by an independent
audit (141 checks, scripts/deep_audit.py).*

1 The observation model

Both tracks share one physical model. A Rydberg atomic receiver measures the magnitude of the total electric field at each sensor. The phase is destroyed by the measurement, which is what makes detection a phase-retrieval problem rather than a linear inverse problem.

1.1 Detection (Track A)

For one channel use, with N atomic sensors and K single-antenna users:

$$\mathbf{z} = |\mathbf{A}^H \mathbf{s} + \mathbf{b} + \mathbf{w}| \quad (1)$$

- $\mathbf{A} \in \mathbb{C}^{K \times N}$ — known channel (Cui eq. 16)
- $\mathbf{s} \in \mathbb{C}^K$ — unknown QAM symbols, $\mathbb{E}|s_k|^2 = 1$
- $\mathbf{b} \in \mathbb{C}^N$ — known local-oscillator reference
- $\mathbf{w} \sim \mathcal{CN}(0, \sigma^2 \mathbf{I})$ — additive noise, inside the magnitude
- $|\cdot|$ is the elementwise amplitude — $|\mathcal{E}|$, never $|\mathcal{E}|^2$

1.2 Channel estimation (Track B)

The unknown becomes the channel and the pilots are known. Over P pilot instants:

$$\mathbf{Z} = |\mathbf{G}\mathbf{S} + \mathbf{B} + \mathbf{W}| \quad (2)$$

with $\mathbf{G} \in \mathbb{C}^{N \times K}$ unknown, $\mathbf{S} \in \mathbb{C}^{K \times P}$ known pilots,

and $\text{vec}(\mathbf{W}) \sim \mathcal{CN}(0, \sigma^2 \mathbf{I})$. This is the exact nonlinear model.

No strong-reference linearisation is used in any Track-B estimator reported here; a runtime tripwire asserts the linearised solver is never called.

1.3 Canonical form

Both reduce to one canonical problem, which is why a single validated solver serves both. Per receive element n :

$$\mathbf{z} = |\mathbf{M}^H \mathbf{u} + \mathbf{b} + \mathbf{w}| \quad (3)$$

	detection	channel estimation
$\mathbf{M}$	$\mathbf{A}$	$\overline{\mathbf{S}}$ (conjugate pilots)
$\mathbf{u}$	$\mathbf{s}$	$\overline{\mathbf{g}}_n$ (conjugate row)
$\mathbf{z}$	$\mathbf{z}$	$\mathbf{Z}_{n, :}$

The channel-estimation adapter loops over receive elements, calls the same solver with

$\mathbf{M} = \mathbf{S}$, $\mathbf{b} = \overline{\mathbf{B}_{n, :}}$, and sets

$\hat{\mathbf{G}}_{n, :} = \overline{\hat{\mathbf{u}}}$. The conjugation was verified by noiseless recovery:

the estimator returns $\mathbf{G}$ to a relative error of 2.4×10^{-15} , and not

$\overline{\mathbf{G}}$.

2 Channel models

2.1 Track A — Cui 3GPP TR 38.901 clustered channel

Cui generates coefficients with the 3GPP TR 38.901 model, giving only Table I. Each user's row is a sum over clusters and rays:

$$a_{n,k} = \sum_{c=1}^{N_c} \sum_{r=1}^{M_r} \alpha_{c,r} e^{-j2\pi f_c \tau_c} (\boldsymbol{\mu}_{eg}^T \boldsymbol{\varepsilon}_{n,c,r}) e^{-j(n-1)\pi \sin \theta_{c,r}} \quad (4)$$

Table I parameter	value
clusters N_c / rays per cluster M_r	23 / 20
path gains $\alpha_{c,r}$	$\mathcal{CN}(0, 1)$
incident angles θ	$\mathcal{U}(-90^\circ, 90^\circ)$
max angle spread per cluster	$\mathcal{U}(-5^\circ, 5^\circ)$
max delay spread	$\mathcal{U}(0, 30 \text{ ns})$
carrier f_c	5 GHz

Rows are normalised so $\text{mean}_n |a_{n,k}|^2 = 1$, which is what makes eqs. (6)–(7) exact. Cui §VI-A samples $\boldsymbol{\varepsilon}_{n,c,r}$ per antenna element; that choice whitens the array response and is the origin of the documented ≈ 2 dB offset in Figs. 5–6 (Section 7).

2.2 Track B — geometric ULA channel

A half-wavelength ULA with resolvable specular paths. For user k :

$$\psi_{\ell,k} = \pi \sin \theta_{\ell,k}, \quad g_{n,k} = \sum_{\ell=1}^{L_k} \alpha_{\ell,k} e^{-j(n-1)\psi_{\ell,k}} \quad (5)$$

equivalently $\mathbf{g}_k = \mathbf{A}(\boldsymbol{\theta}_k) \boldsymbol{\alpha}_k$ with the array manifold

$\mathbf{a}(\theta) = [1, e^{-j\psi}, \dots, e^{-j(N-1)\psi}]^T$, $\|\mathbf{a}\|_2^2 = N$.

Parameters: $\theta_{\ell,k} \sim \mathcal{U}[-\pi/2, \pi/2]$,

$\alpha_{\ell,k} \sim \mathcal{CN}(0, \beta_k/L_k)$ so that $\mathbb{E}[g_{n,k}|^2] = \beta_k$, and

$L_k \sim \mathcal{U}\{3, \dots, 7\}$ drawn independently per user per realisation.

3 SNR and RSR calibration

These two definitions set every operating point in every figure. Both were re-derived from first principles during the audit and checked against the implementation.

3.1 SNR — Cui eq. (36)

$$\text{SNR} = \frac{\mathbb{E}(|\mathbf{a}_n^H \mathbf{s}|^2)}{\mathbb{E}(|\mathbf{w}_n|^2)} \quad (6)$$

With row-normalised $\mathbf{A}$, unit-energy QAM and independent users, the numerator is $\sum_k \mathbb{E}|a_{n,k}|^2 \mathbb{E}|s_k|^2 = K$. Hence the noise variance actually used is

$$\sigma^2 = \frac{K}{\text{SNR}_{\text{lin}}}, \quad \text{SNR}_{\text{lin}} = 10^{\text{SNR}_{\text{dB}}/10} \quad (7)$$

Note this is total signal power over all K users, so fixing the SNR does not fix the per-user SNR: doubling K at fixed SNR doubles σ^2 .

3.2 RSR — Cui eq. (37)

$$\text{RSR} = \frac{\mathbb{E}(|b_n|^2)}{\mathbb{E}(|a_{n,k} s_k|^2)} \quad (8)$$

The denominator is a **singleuser**'s contribution, not the sum over K . This is the single easiest place to introduce a factor- K error. With $\beta_{\text{ref}} = 1$ and $\mathbb{E}|s_b|^2 = 1$ it gives

$$|\alpha_b| = \sqrt{\text{RSR}_{\text{lin}}} \quad (\text{not } \sqrt{K \text{RSR}_{\text{lin}}}, \text{ not } \sqrt{\text{RSR}_{\text{lin}}/K}) \quad (9)$$

The audit explicitly tests that the implemented value differs from both incorrect alternatives, and measures the achieved SNR and RSR empirically from generated worlds: 2.82 dB and 12.15 dB against targets of 3 and 12 dB.

4 Detection and estimation algorithms

4.1 Spectral initialisation

All iterative solvers start from Cui's augmented spectral initialiser. With $\tilde{\mathbf{m}}_q = [\mathbf{m}_q; b_q]$:

$$\mathbf{M}_{\text{spec}} = \sum_q z_q \tilde{\mathbf{m}}_q \tilde{\mathbf{m}}_q^H \in \mathbb{C}^{(D+1) \times (D+1)}, \quad \tilde{\mathbf{u}}_0 = \text{principaleigenvector} \quad (10)$$

4.2 Biased Gerchberg-Saxton (Cui Algorithm 1)

GS alternates between restoring the measured magnitude with the currently estimated phase, and re-solving a linear least squares. At iteration t :

$$\boldsymbol{\lambda}^{t-1} = \mathbf{M}^H \mathbf{u}^{t-1} + \mathbf{b}, \quad \boldsymbol{\theta}^t = \angle \boldsymbol{\lambda}^{t-1} \quad (11)$$

$$\mathbf{y}^t = \mathbf{z} \odot e^{j\boldsymbol{\theta}^t}, \quad \mathbf{r}^t = \mathbf{y}^t - \mathbf{b} \quad (12)$$

$$(\mathbf{M}\mathbf{M}^H)\mathbf{u}^t = \mathbf{M}\mathbf{r}^t \quad (13)$$

The measured amplitude $\mathbf{z}$ is kept exactly; only the phase is supplied by the current iterate. Solved via the normal equations, never an explicit inverse. Cui sets $t_0 = 50$ iterations, as used throughout.

4.3 EM-GS (Cui Algorithm 2)

EM-GS keeps the same phase and least-squares steps but weights the restored observation by the Bessel ratio, which is the conditional mean of the Rician amplitude — a soft, SNR-aware version of the hard magnitude substitution:

$$R(\kappa) = \frac{I_1(\kappa)}{I_0(\kappa)}, \quad \boldsymbol{\kappa} = \frac{2}{\sigma^2} \mathbf{z} \odot |\boldsymbol{\lambda}| \quad (14)$$

R is monotone increasing with $R(0) = 0$ and $R(\kappa) \rightarrow 1$. That limit explains an observed behaviour: at high SNR or high RSR $R \rightarrow 1$ and EM-GS degenerates into plain GS, so the two become indistinguishable. The implementation evaluates R with exponentially scaled Bessel functions to avoid overflow, never as a raw ratio I_1/I_0 . The audit checks it against SciPy and against the quadrature definition

$$I_n(\kappa) = \frac{1}{\pi} \int_0^\pi e^{j\kappa \cos t} \cos(nt) dt.$$

4.4 Genie ZF with known phase

Given the true noisy phase — information a magnitude-only receiver never has — the problem becomes linear:

$$\mathbf{r} = \mathbf{z} \odot e^{j\boldsymbol{\theta}} - \mathbf{b}, \quad (\mathbf{M}\mathbf{M}^H)\hat{\mathbf{u}} = \mathbf{M}\mathbf{r} \quad (15)$$

This is a benchmark, not a valid estimator. It is allowed to lie below the CRLB of Section 4.6, because that bound applies to estimators seeing only $\mathbf{z}$.

4 Detection and estimation algorithms (cont.)

4.5 Exhaustive search — LS and ML

Cui's two optimal-detector benchmarks, searched over the full constellation to the power K :

$$J_{\text{LS}}(\mathbf{u}) = \|\mathbf{z} - \mathbf{M}^H \mathbf{u} + \mathbf{b}\|_2^2 \quad (16)$$

$$J_{\text{ML}}(\mathbf{u}) = \sum_q \left[-\frac{|\lambda_q|^2}{\sigma^2} + \log I_0\left(\frac{2z_q |\lambda_q|}{\sigma^2}\right) \right] \quad (17)$$

LS and ML are not assumed identical: (17) is the $\mathbf{u}$ -dependent part of the exact Rician log-likelihood, evaluated in the log domain. Feasible for Fig. 7(a)/8 ($4^3 = 64$ candidates) but not for Fig. 7(b) ($16^6 \approx 1.7 \times 10^7$), which is exactly why Cui omits it there and so do we.

4.6 Cramér-Rao lower bound

Each amplitude is Rician with density

$$p(z | \lambda) = \frac{2z}{\sigma^2} \exp\left(-\frac{z^2 + |\lambda|^2}{\sigma^2}\right) I_0\left(\frac{2z|\lambda|}{\sigma^2}\right) \quad (18)$$

giving the Fisher information and bound

$$\mathbf{F} = \sum_q \beta_q \mathbf{m}_q \mathbf{m}_q^H, \quad \beta_q = \frac{\mathbb{E}[z_q^2 R^2(\kappa_q)] - |\lambda_q|^2}{\sigma^4} \quad (19)$$

β_q is evaluated by numerical quadrature over a two-sided window centred on $|\lambda|$, never clipped to its high-SNR limit. As $\text{SNR} \rightarrow \infty$, $\beta_q \rightarrow 1/(2\sigma^2)$, so

$\mathbf{F} \rightarrow \frac{1}{2\sigma^2} \mathbf{M} \mathbf{M}^H$ and the bound approaches

$2\sigma^2 (\mathbf{M} \mathbf{M}^H)^{-1}$ — exactly $10 \log_{10} 2 = 3.0103$ dB above the genie-ZF

covariance. That is the price of losing phase, and the audit measures it at 3.0103 dB.

5 Metrics and aggregation

How results are pooled matters as much as how they are computed. Two rules are applied everywhere and are the most common source of silently wrong curves.

5.1 NMSE — ratio of sums, never mean of ratios

$$\text{NMSE} = \frac{\sum_{\text{trials}} \|\mathbf{s} - \tilde{\mathbf{s}}\|_2^2}{\sum_{\text{trials}} \mathbb{E} \|\mathbf{s}\|_2^2}, \quad \text{NMSE}_{\text{dB}} = 10 \log_{10} (\text{NMSE}) \quad (20)$$

For channel estimation the same rule uses Frobenius energies,

$\text{NMSE}_G = \sum \|\hat{\mathbf{G}} - \mathbf{G}\|_F^2 / \sum \|\mathbf{G}\|_F^2$. Three points matter:

- It is $10 \log_{10}$, not $20 \log_{10}$ — NMSE is already a power ratio, so a factor-2 dB error is the trap here.
- The denominator is the *expected* symbol energy K for unit-energy QAM, not a per-trial $\|\mathbf{s}\|^2$.
- $\tilde{\mathbf{s}}$ is the *continuous* solver output, before any constellation demapping.

Averaging per-trial dB values instead would give a different and smaller number, because it is a geometric rather than arithmetic mean of the energies. The audit constructs a case where the two differ by 7 dB.

5.2 BER — global bit errors over global bits

The demapping step Cui describes: project each recovered symbol to the nearest constellation point, then to bits.

$$\text{BER} = \frac{\sum_{\text{trials}} (\text{biterrors})}{\sum_{\text{trials}} (\text{bitcount})} \quad (21)$$

Not the mean of per-trial BERs. Gray mapping means adjacent constellation points differ in exactly one bit, so a symbol error usually costs one bit of $\log_2 M$; consequently $\text{SER} / \log_2 M \leq \text{BER}$. The audit verifies the Gray property by walking every constellation neighbour ($M = 4, 16, 64$, zero violations) and recounts the bit errors independently from the constellation bit table.

5.3 Zero-error points

At high SNR some points produce no bit errors at all. A zero cannot be drawn on a log axis and must not be reported as $\text{BER} = 0$. We use the one-sided Wilson score bound:

$$\text{BER} \leq \frac{\hat{p} + \frac{z^2}{2n} + z \sqrt{\frac{\hat{p}(1-\hat{p})}{n} + \frac{z^2}{4n^2}}}{1 + \frac{z^2}{n}}, \quad z = 1.645 \quad (22)$$

which stays valid at $k = 0$ and for small n , unlike the normal approximation. Such points are omitted from the plotted curve and their bounds retained in the aggregate CSV. Section 7 shows these bounds already sit below anything Cui plots, so the resolution is sufficient.

5.4 Common random numbers

At each operating point one frozen world $\{\mathbf{A}, \mathbf{s}, \mathbf{b}, \mathbf{w}, \mathbf{z}\}$ is generated from a key derived from (trial, SNR, RSR) and handed to every algorithm. Comparisons are therefore paired, removing channel variability from algorithm differences, and any trial is reproducible in isolation.

6 Track A — reproduction of Cui's figures

Five published figures, all on the Cui channel with the equations of Sections 3–5 and $t_0 = 50$ iterations. Error bars are deliberately not drawn; confidence intervals live in the aggregate CSVs.

6.1 Fig. 5 — detection NMSE vs SNR

$N \times K = 36 \times 3$, 16-QAM, RSR = 12 dB, 2 000 trials/SNR, 18 points. Uses eqs. (1), (4), (6)–(7), (10)–(14), (15), (19), (20).

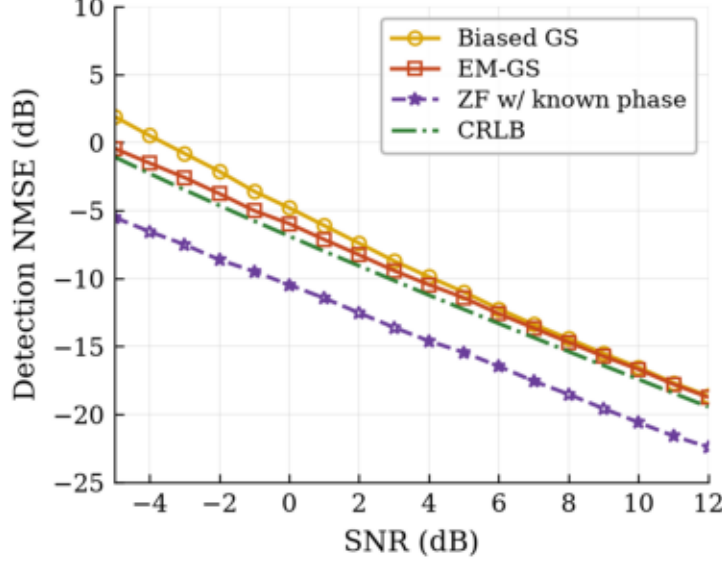


Fig. 5 — detection NMSE vs SNR. Every ordering Cui reports is reproduced: EM-GS below biased GS at all SNR, both approaching the CRLB, and the CRLB sitting 3.01 dB above genie ZF at high SNR.

The absolute level sits about 2 dB below Cui's published curve. This is a channel-conditioning effect, discussed in Section 7, not a solver or calibration error.

6.2 Fig. 6 — detection NMSE vs RSR

$N \times K = 36 \times 3$, 16-QAM, SNR = 3 dB, 500 trials/RSR, 26 points, RSR 0...25 dB.

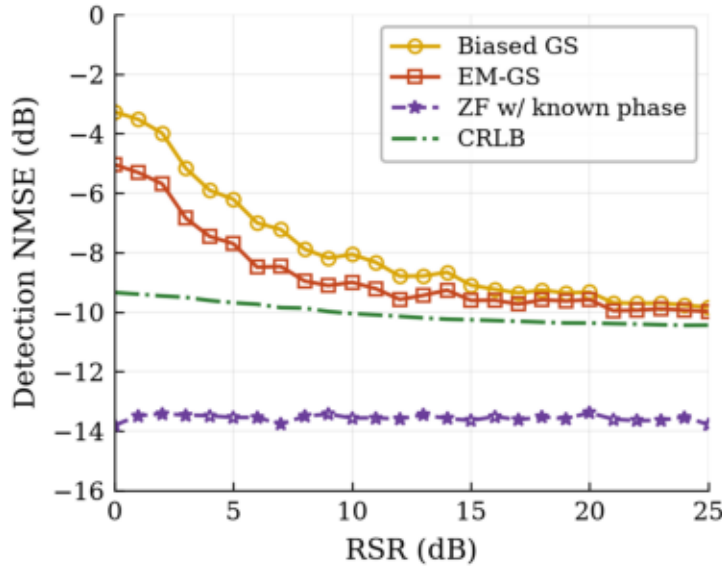


Fig. 6 — detection NMSE vs RSR. The genie-ZF curve is flat because its error $(\mathbf{A}\mathbf{A}^H)^{-1}\mathbf{A}\mathbf{w}$ does not involve $\mathbf{b}$; the fitted slope is -0.0026 dB/dB, not statistically significant.

The GS/EM-GS gap shrinks as RSR grows, from -1.77 dB at $\text{RSR} = 0$ to -0.14 dB at $\text{RSR} = 25$, which is exactly the $R(\kappa) \rightarrow 1$ behaviour predicted by eq. (14).

6 Track A — reproduction of Cui's figures (cont.)

6.3 Fig. 7(a) — BER vs SNR, small-scale

$N \times K = 36 \times 3$, 4-QAM, RSR = 12 dB. 333 000 trials, 1 998 000 bits per algorithm. Adds eqs. (16), (17), (21), (22).

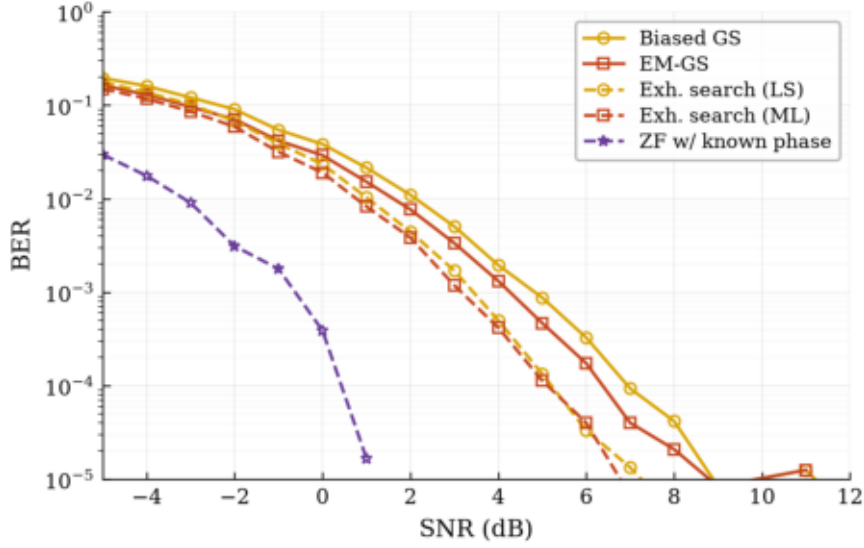


Fig. 7(a) — BER vs SNR. Ordering matches Cui exactly: ZF, then exhaustive ML below exhaustive LS, then EM-GS below biased GS, each dashed optimal search below its iterative counterpart.

6.4 Fig. 7(b) — BER vs SNR, large-scale

$N \times K = 100 \times 6$, 16-QAM, RSR = 12 dB. 122 000 trials, 2 928 000 bits per algorithm. Exhaustive search omitted exactly as Cui does, since $16^6 \approx 1.7 \times 10^7$ candidates per trial is prohibitive.

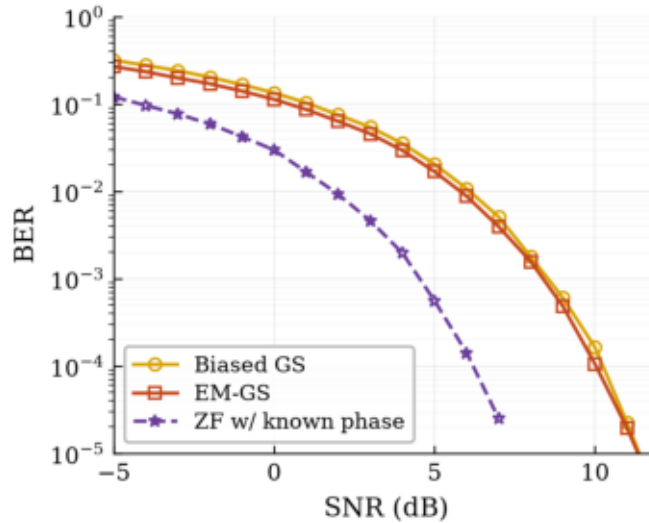


Fig. 7(b) — BER vs SNR. The measured EM-GS-to-ZF SNR gap is 3.79–3.97 dB across BER 3×10^{-2} to 10^{-3} ; Cui states "between 3 ~ 4 dB".

6 Track A — reproduction of Cui's figures (cont.)

6.5 Fig. 8 — BER vs RSR

$N \times K = 36 \times 3$, 4-QAM, SNR = 3 dB, RSR 0...25 dB. 239 000 trials, 1 434 000 bits per algorithm.

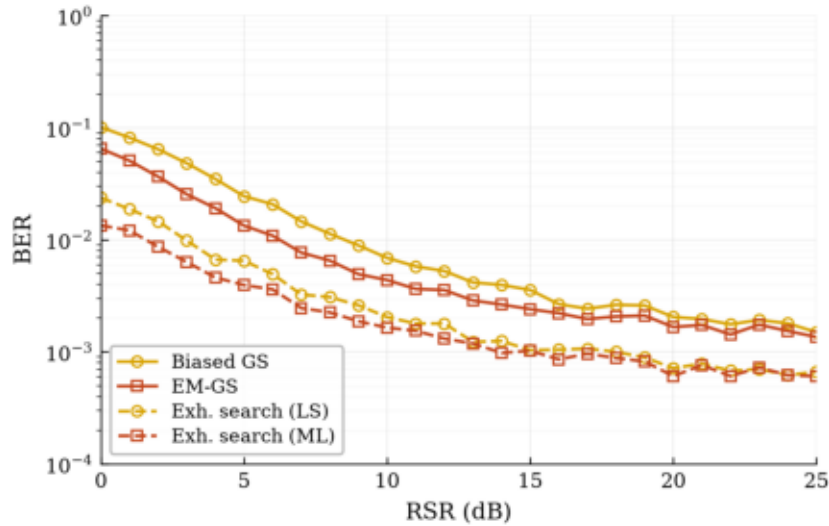


Fig. 8 — BER vs RSR. Cui plots no genie-ZF curve here, so neither do we, though it is evaluated and kept in the CSV.

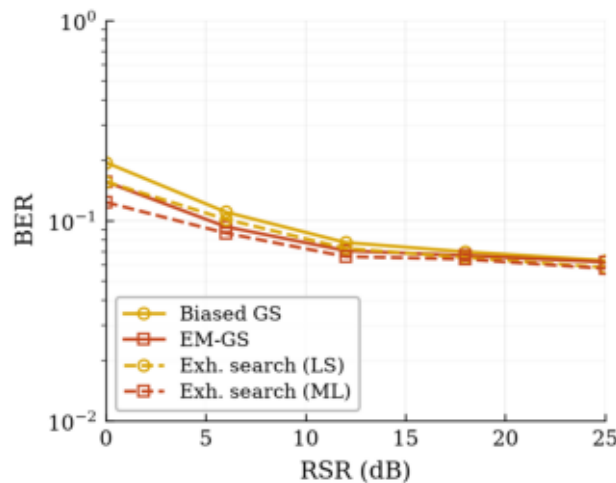
Measured EM-GS improvement from RSR 0 to 20 dB is $39.1 \times$; Cui states "more than one order of magnitude".

The caption/body contradiction in Cui's Fig. 8

Cui's Fig. 8 caption says 16-QAM; the body text says "the SNR is fixed as 3 dB and a 4-QAM modulator is adopted". The two cannot both be right, so both were run and compared against the published levels over 12 matched (algorithm, RSR) points.

	4-QAM (body text)	16-QAM (caption)
median BER ratio to Cui	0.82	24.12
median $ \log_{10} \text{ ratio} $	0.086	1.367

The 16-QAM variant sits about $29 \times$ above the 4-QAM curve and misses Cui's levels by more than an order of magnitude beyond RSR = 0. The body text is correct and the caption is in error.



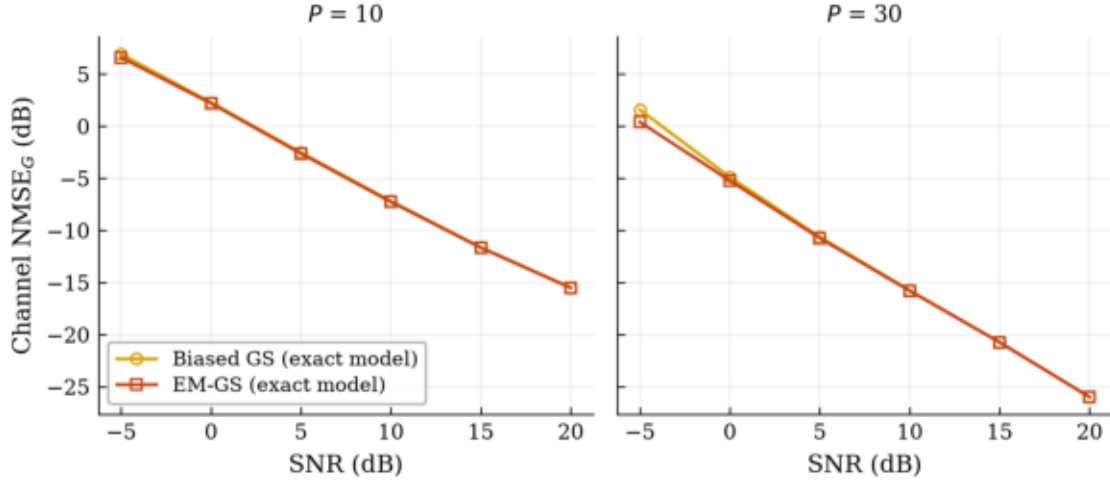
Diagnostic only — Fig. 8 rerun at 16-QAM as the caption claims. Note the vertical scale: these levels are far from the published figure.

7 Track B — exact-model ULA channel estimation

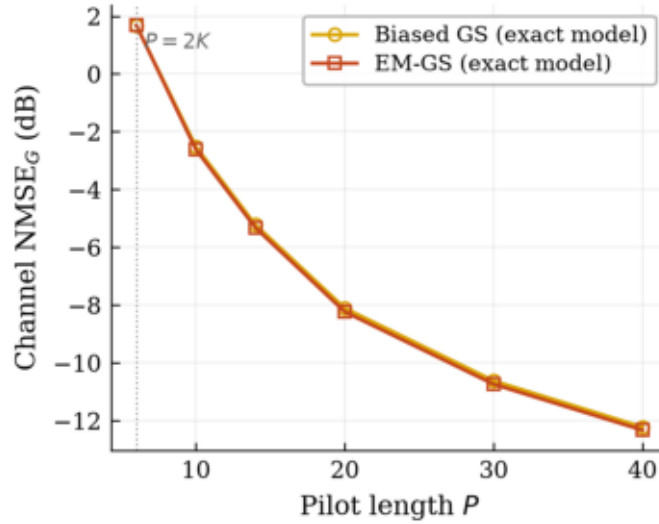
Track B asks a different question: can the geometric structure of a ULA channel improve estimation while keeping Cui's exact nonlinear magnitude model? The baselines below are Cui's own algorithms, unmodified, applied to the ULA channel of eq. (5).

7.1 Baselines B1 and B2

$N = 8$ (matching Xu's $I = 8$ for comparability), $K = 3$, $L_k \sim \mathcal{U}\{3, \dots, 7\}$ per realisation, RSR = 12 dB, $t_0 = 50$, 400 trials/point. Metric is eq. (20) in its Frobenius form.



B1 — channel NMSE vs SNR at two pilot lengths. Both estimators operate on the exact model $\mathbf{Z} = [\mathbf{GS} + \mathbf{B} + \mathbf{W}]$; no linearisation is involved.



B2 — channel NMSE vs pilot length at SNR = 5 dB. $P = 2K$ is the counting bound below which the problem is underdetermined.

GS and EM-GS are nearly indistinguishable here because RSR = 12 dB already drives $R(\kappa) \rightarrow 1$. The measured separation grows as RSR falls: +1.04 dB at RSR = 0, +0.17 dB at 12, +0.01 dB at 30.

7 Track B — structural estimation (cont.)

7.2 The structural constraint

Equation (5) says each channel column is a sum of L_k complex exponentials. By Kronecker's theorem, a length- N sequence is such a sum of L exponentials if and only if its Hankel matrix has rank L . That gives an exact algebraic characterisation of the model — no grid, and the angles and gains never appear:

$$\min_{\mathbf{G}} \|\mathbf{Z} - |\mathbf{G}\mathbf{S} + \mathbf{B}|\|_F^2 \quad \text{s. t.} \quad \text{rank } \mathcal{H}(\mathbf{g}_k) \leq L_k \quad (23)$$

Three projections onto this set were implemented and compared. Cadzow was chosen because it is the only one that is simultaneously grid-free and a genuine alternating projection between two closed sets with exact projectors — SVD truncation (Eckart-Young) and anti-diagonal averaging onto the Hankel subspace. Angular OMP is grid-based and leaves about 6.9 % residual even on noiseless structured data; ESPRIT is grid-free but is a parameter estimator, not idempotent, with no variational characterisation.

7.3 Why structure must be enforced inside the iteration

Cui's row adapter is separable across receive elements, so an unstructured sweep cannot see a coupling along n . Applying the projection once and then running GS to convergence achieves nothing, because the fixed points of $T_{\text{GS}}^\infty \circ P_S$ are just the fixed points of T_{GS} .

Measured: the gain decays monotonically with the number of unconstrained iterations after projection — +1.30 dB at 1, +0.06 at 10, exactly 0.00 at 50. Interleaving instead gives the iteration map

$$T = P_S \circ T_{\text{GS}} \quad (24)$$

whose fixed points are simultaneously structured and consistent with the measurement update. The measurement step remains exactly eqs. (11)–(14) on $\mathbf{Z}$.

7.4 Identifiability — when structure can help

A length- N Hankel matrix has rank at most $\lfloor N/2 \rfloor$. At $N = 8$ that cap is 4, so for $L_k \geq 5$ — 60 % of the $\mathcal{U}\{3..7\}$ prior — the true channel is already full rank and the constraint is vacuous. Counting real degrees of freedom, the unstructured model has $2NK$ and the geometric one $3\sum_k L_k$:

N	unstructured $2NK$	geometric $3\sum_k L_k$	reduction
8	48	45	1.07×
16	96	45	2.13×
32	192	45	4.27×

So at the frozen $N = 8$ the prior carries almost no information, and measured gains are mixed (+0.92 dB at low SNR, −1.48 dB at high). At $N = 16$ and $N = 32$ the constraint becomes active and the gains turn consistently positive — up to +3.86 dB at $N = 32$, where the structured estimator won every one of 40 trials. This is a property of the configuration, not of the algorithm.

8 Verification

Every formula above was checked numerically by re-deriving it independently rather than reading the implementation and agreeing with it. 141 checks, no blocker, high or medium findings.

quantity	independent reference	result
$\sigma^2 = K/\text{SNR}$	derived from eq. (6)	exact
$ \alpha_b $	eq. (8); ruled out $\sqrt{K \cdot \cdot}$, $\sqrt{\cdot/K}$	exact
$\mathbb{E} w ^2$	empirical over 400 worlds	ratio 1.000
$R(\kappa)$	SciPy and quadrature of I_n	$< 10^{-9}$
Gray mapping	walked all neighbours, $M = 4, 16, 64$	0 violations
bit errors	recounted from constellation table	exact match
CRLB / ZF gap	analytic $10\log_{10} 2$	3.0103 dB
conjugation	noiseless recovery of $\mathbf{G}$	2.4×10^{-15}
modulation per figure	bits/trial = $K\log_2 M$	confirmed
linearisation leak	runtime tripwire on HS-GS	0 calls
store integrity	6 stores: fingerprint, duplicates	0 duplicates

8.1 Agreement with Cui

Curves were extracted from the published PDF by colour segmentation with axis calibration, then compared both vertically (BER ratio) and horizontally (the SNR shift aligning the curves, which is the meaningful measure for a waterfall).

figure	qualitative	absolute	discrepancy
Fig. 5	full	offset	≈ 2 dB
Fig. 6	full	offset	≈ 2 dB
Fig. 7(a)	full	yes	-0.90 to $+0.05$ dB
Fig. 7(b)	full	yes	-0.10 to -0.15 dB
Fig. 8	full	yes	-0.80 to -1.40 dB

Two quantities Cui states in prose were reproduced without any curve extraction at all, which is the strongest available evidence since it bypasses the pixel reading entirely: the Fig. 7(b) EM-GS-to-ZF gap (stated 3–4 dB, measured 3.79–3.97) and the Fig. 8 improvement from RSR 0 to 20 dB (stated more than one order of magnitude, measured $39.1 \times$).

8.2 The residual 2 dB in Figs. 5 and 6

The offset is constant across both NMSE sweeps, which is the signature of channel statistics rather than of a solver, calibration or metric error — and the BER figures, which use the same channel, solvers and calibration, agree to within 0.1–1.4 dB. Two audits against the actual TR 38.901 specification found no zero-free-parameter correction that closes it. The cause is that Cui's own Table I fixes the angle, delay and cluster statistics that would otherwise pin the channel down, while leaving the array geometry and the orientation of μ_{eg} unstated. A sweep showed the achievable shift spans 0.0–1.3 dB purely by rotating that unspecified frame, so the residual is not identifiable from the published information.

8.3 What could not be reproduced

- CM-ZF — described only as extending a cited approximation to the biased phase-retrieval problem. That is not a specification, so it was deliberately not invented, and appears in no figure.
- The absolute NMSE level of Figs. 5–6, for the reason in Section 8.2.

- Fig. 8's intended modulation, which the paper itself states inconsistently; resolved empirically in Section 6.5.